VENTRICULAR SYSYTEM

Ventricular systen

COMPONENTS:

There are 4 ventricles in the brain filled with CSF(cerebrospinal fluid).

- **2 lateral venricles** *(in brain hemispheres)*
- **3rd ventricle** *(in diencephalon)*
- **4th ventricle** *(between the pons, medulla and cerebellum)*

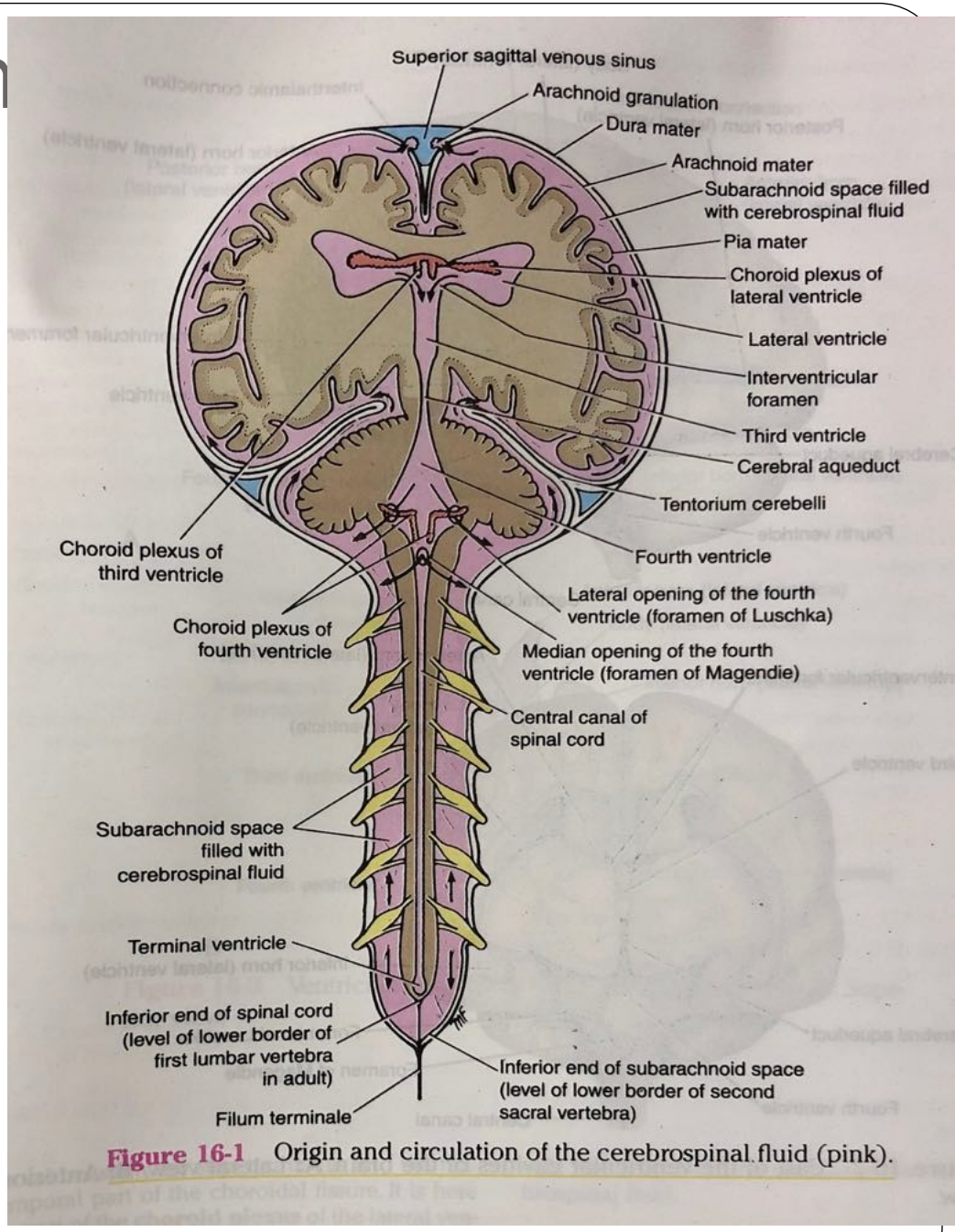
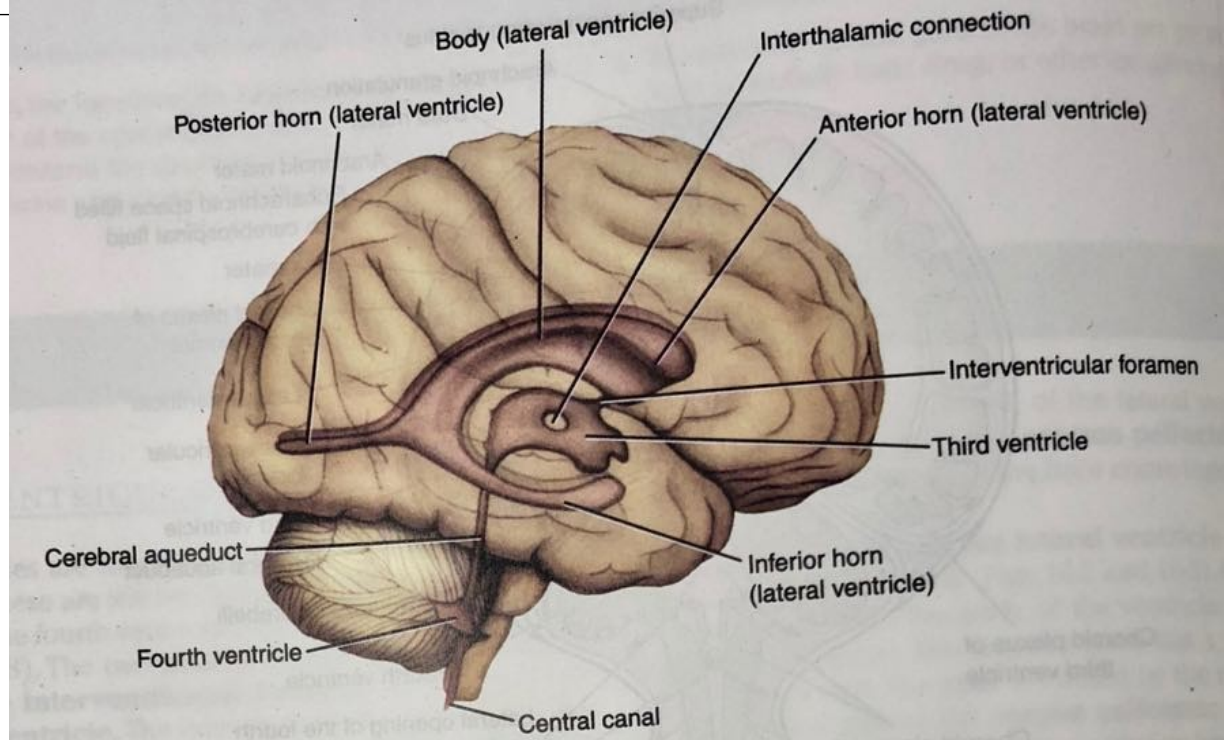
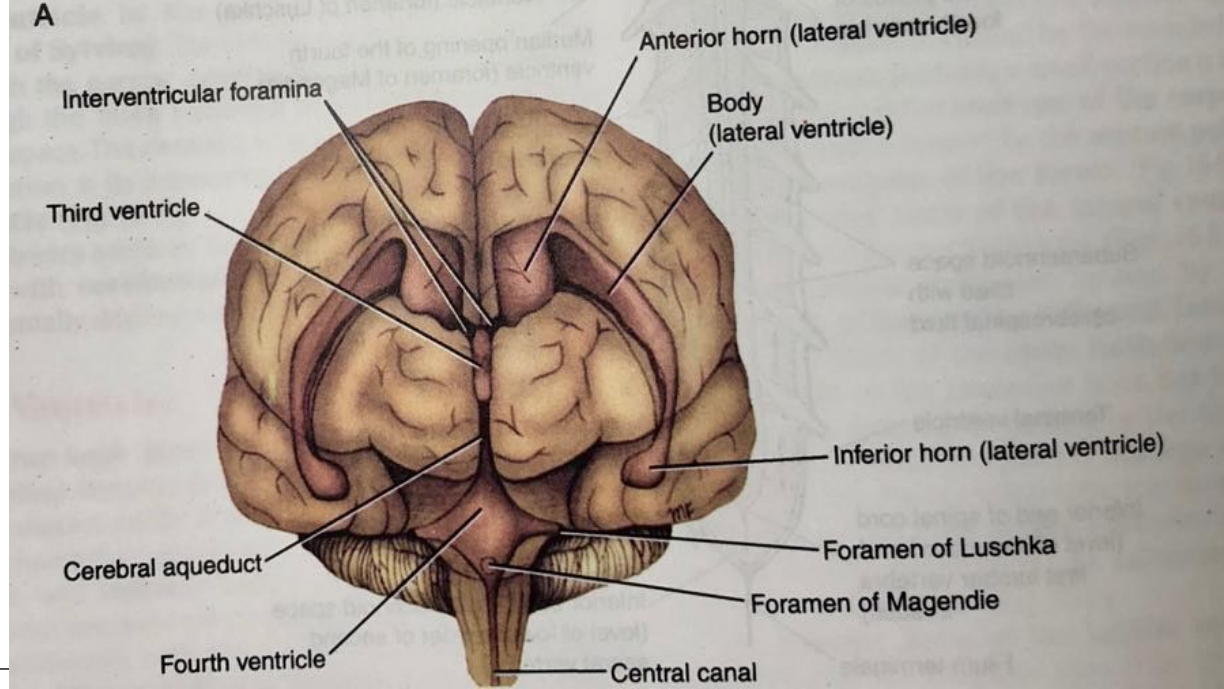


Figure 16-1 Origin and circulation of the cerebrospinal fluid (pink).

These ventricles are connected by:

- **The foramen of Monro (interventricular foramen):**
  lateral ventricles >> 3rd ventricle
- **Cerebral aqueduct (Aqueduct of Sylvius)**
  3rd ventricle >> 4th ventricle
- **The foramen of magendie and lushka**
  4th ventricle >> sub-arachnoid space





Lateral ventricles

- Two large ventricles, one is present in each cerebral hemisphere.
- C-shaped cavity
- It has a:
  - Body: in parietal lobe
  - Anterior horn: in frontal lobe
  - Posterior horn: in occipital lobe
  - Inferior horn: in temporal lobe

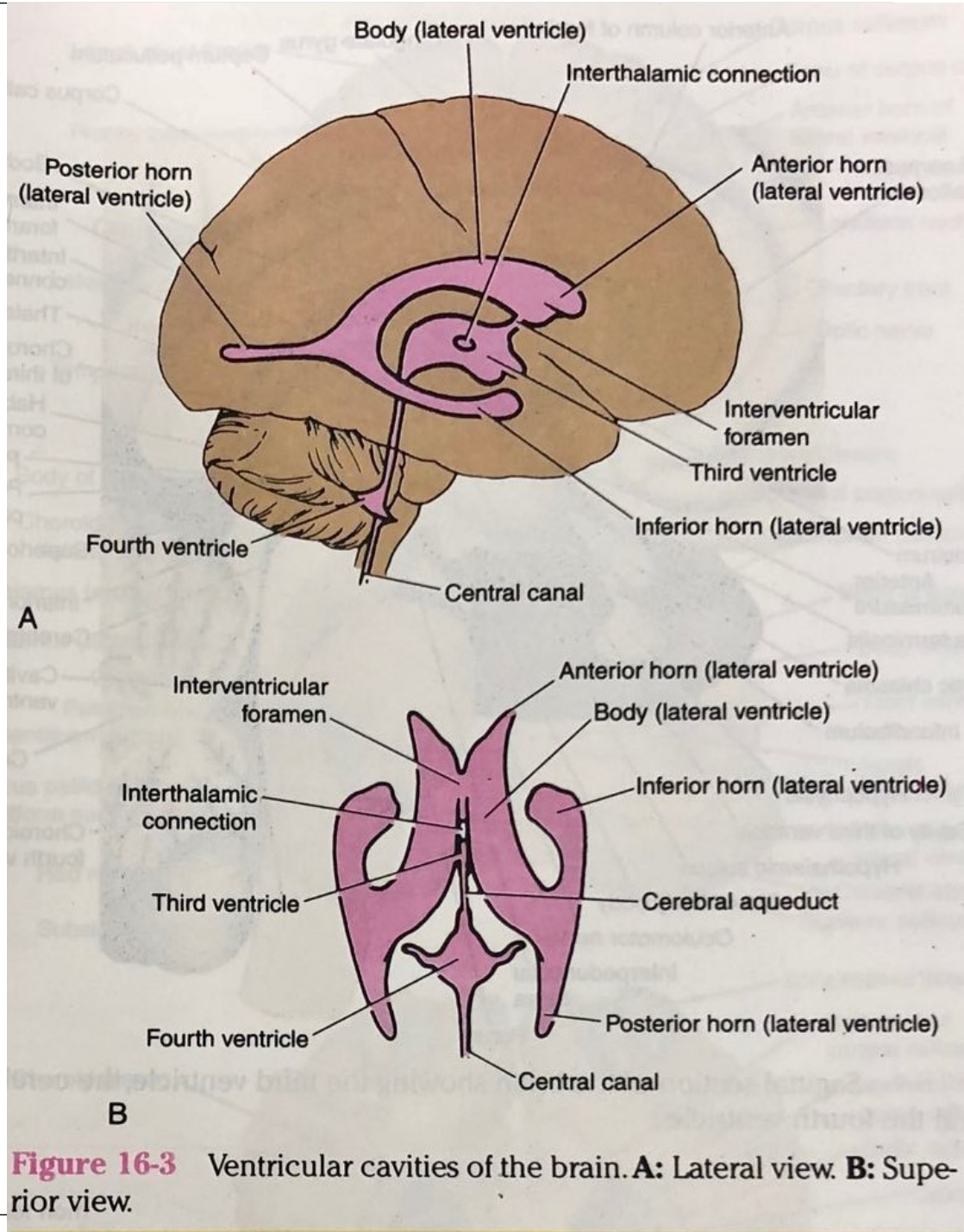


Figure 16-3 Ventricular cavities of the brain. **A:** Lateral view. **B:** Superior view.

Body of lateral ventricle

- Lies in parietal lobe

ROOF:

Undersurface of corpus callosum

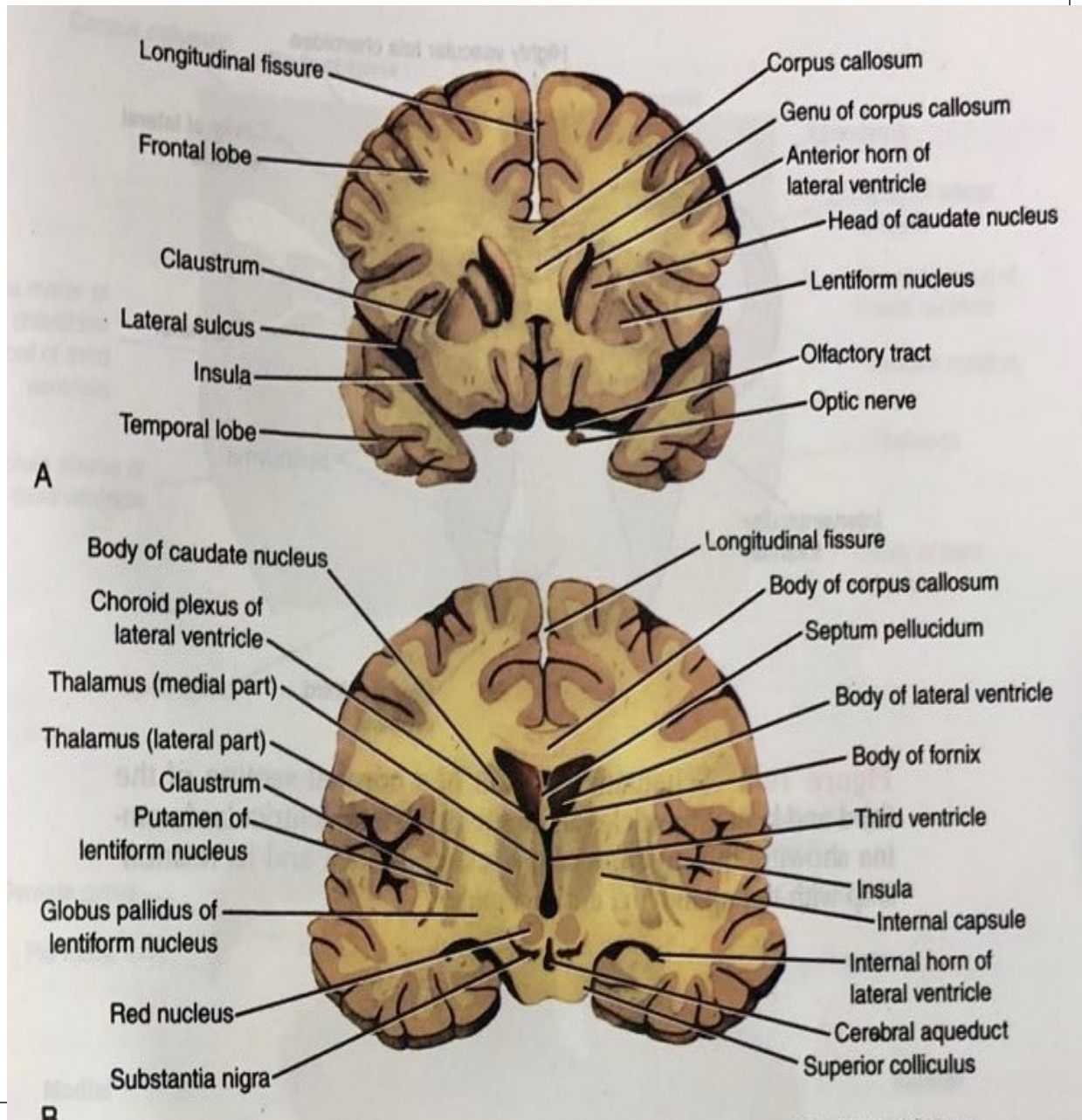
FLOOR:

Lateral to medial it is formed by:

- Body of caudate nucleus
- Upper surface of thalamus
- Choroid plexus
- Body of fornix

MEDIAL WALL:

- *Anteriorly:* Septum pellucidum
- *Posteriorly:* roof and floor come together



Anterior horn

- It is in the frontal lobe.
- **Boundaries:**

ROOF:

Under surface of Corpus callosum

FLOOR:

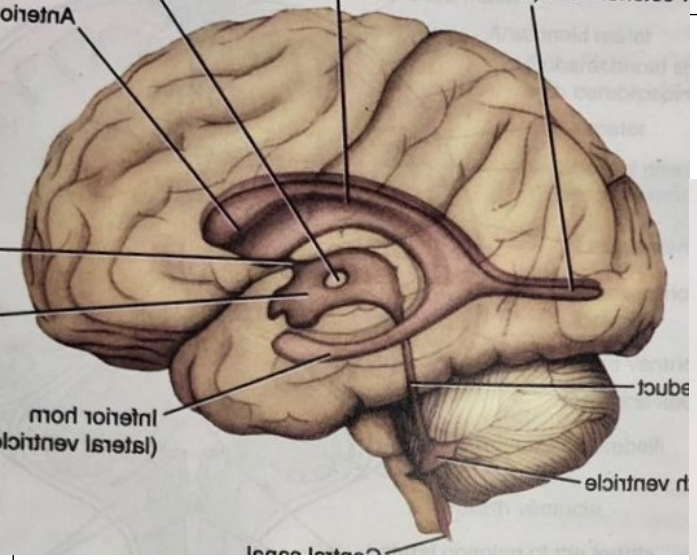
Rostrum of corpus callosum

ANTERIORLY:

Genu of corpus callosum

MEDIALLY:

Septum pellucidum



ROOF:
Under surface of Corpus callosum

FLOOR:
Rostrum of corpus callosum

ANTERIORLY:
Genu of corpus callosum

MEDIALLY:
Septum pellucidum

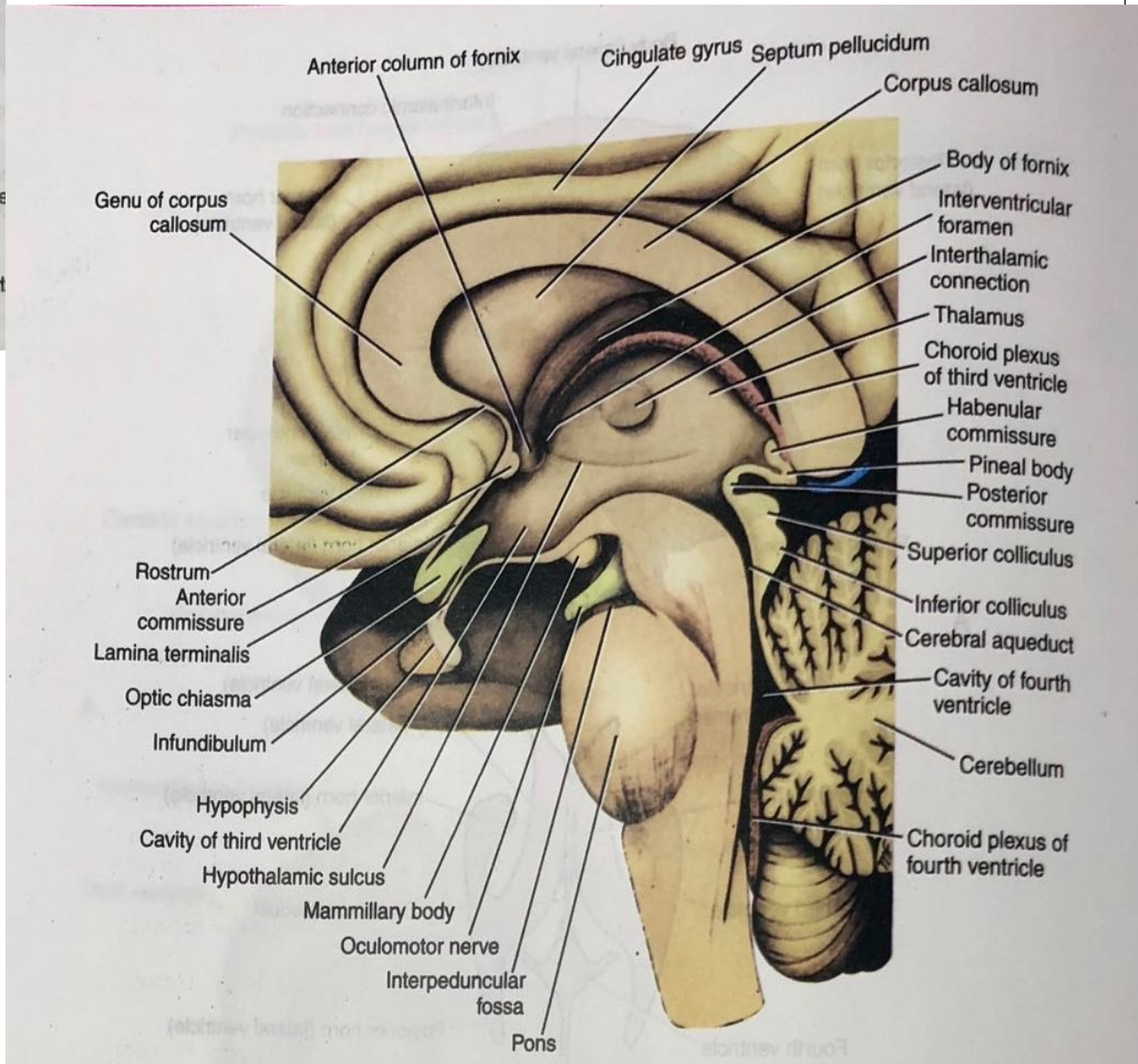


Figure 16-4 Sagittal section of the brain showing the third ventricle, the cerebral aqueduct, and the fourth ventricle.

Posterior horn

- Lies in the occipital lobe.

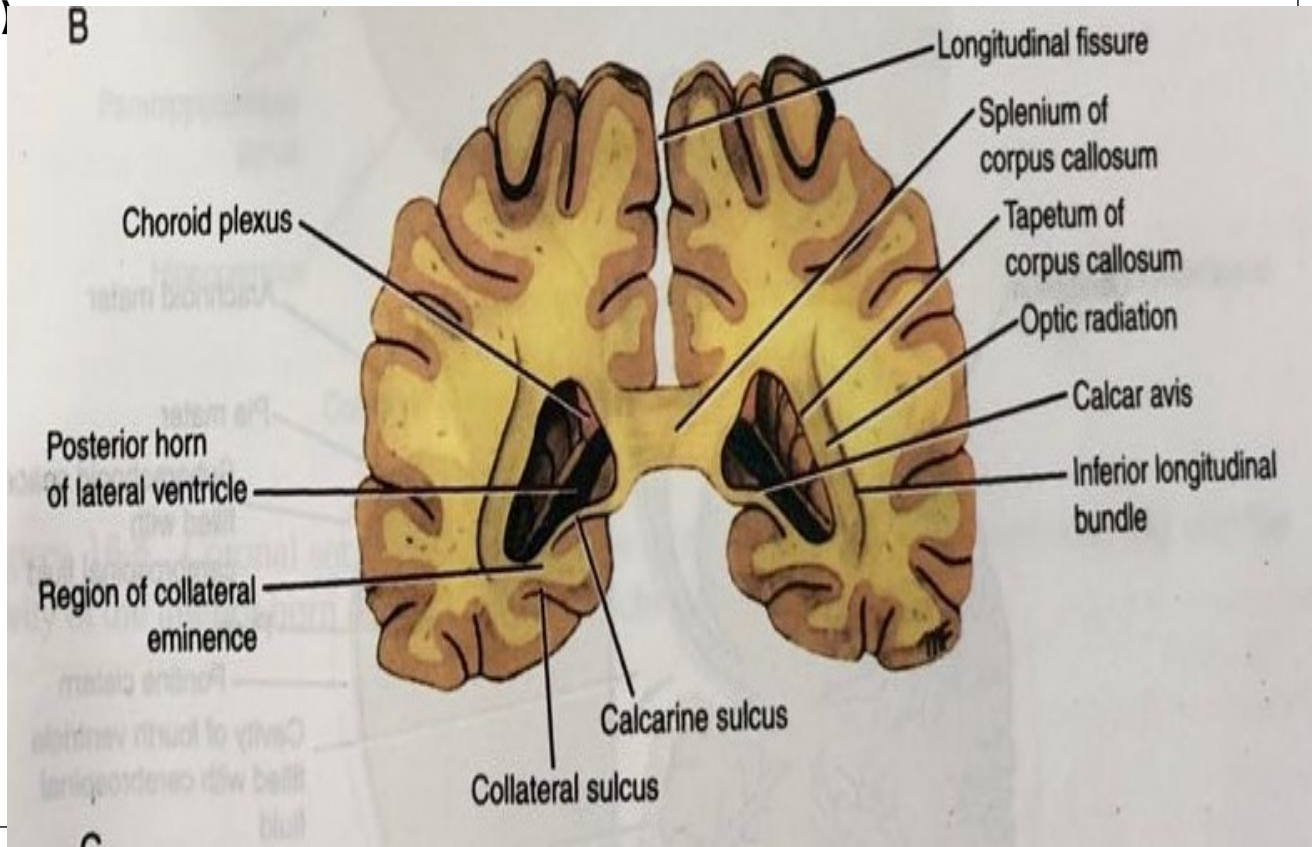
ROOF, LATERAL WALL:

Formed by the Tapetum of the corpus callosum

MEDIALLY:

There are two elevations

- Superior swelling- **(Bulb of superior horn)**
  formed by forceps major
- Inferior swelling- **(Calcar Avis)**
  Formed by Calcarine sulcus



Inferior horn

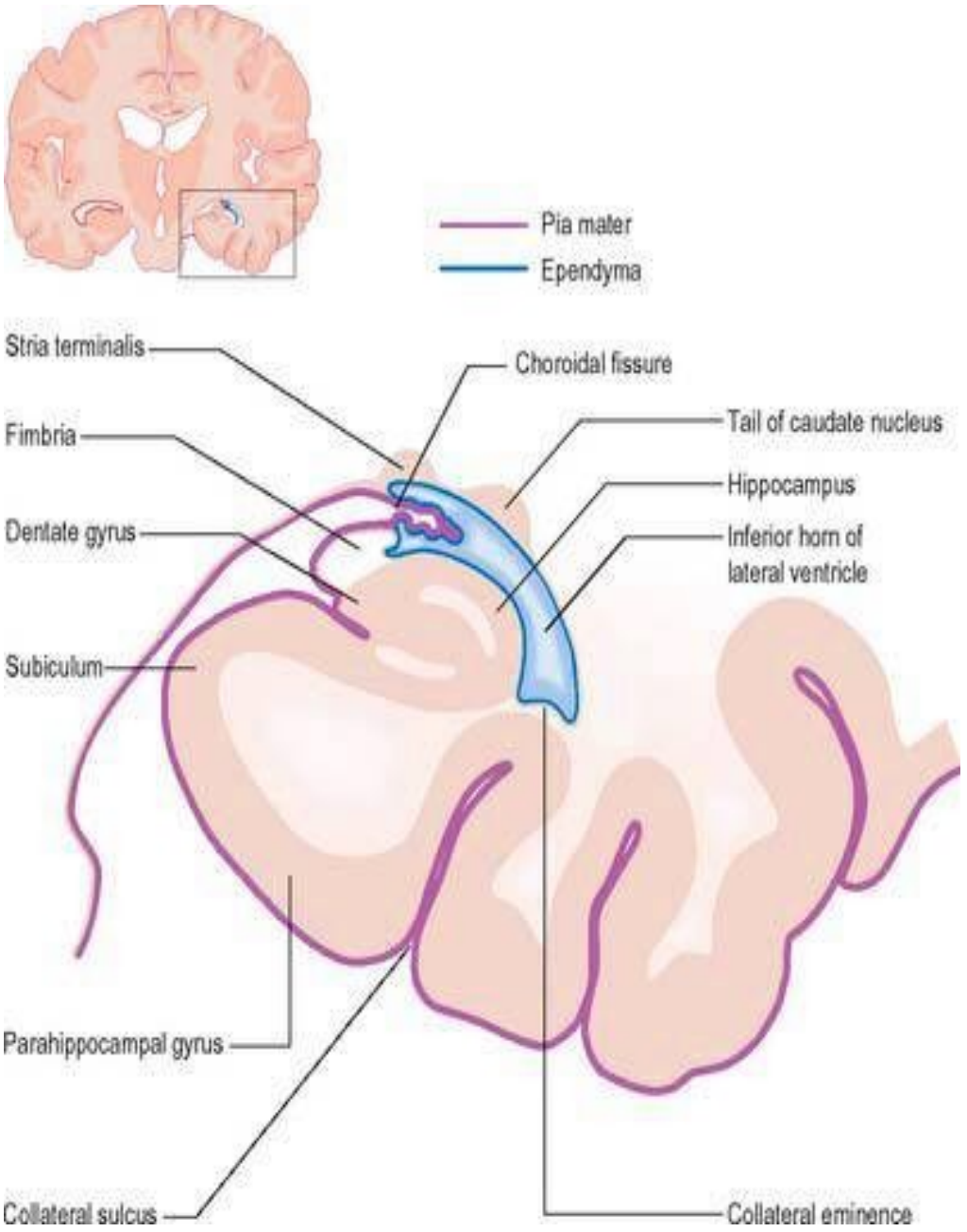
- It lies in the temporal lobe.

ROOF:

- Tapetum of corpus callosum
- Tail of caudate nucleus
- Amygdaloid nucleus
- Stria terminalis

FLOOR:

- Collateral eminence
- Hippocampus
- Fimbria of hippocampus



3rd ventricle

- It is cavity within Diencephalon
- It is a midline slit-like cavity situated between the two thalami and the part of hypothalamus.

BOUNDARIES:

- **Anterior wall:**
  from above downwards by
  - Anterior column of fornix
  - Anterior commissure
  - Lamina terminalis

- **Posterior wall:**
  From above downwards by
  - Pineal gland
  - Posterior commissure
  - Commencement of cerebral aqueduct

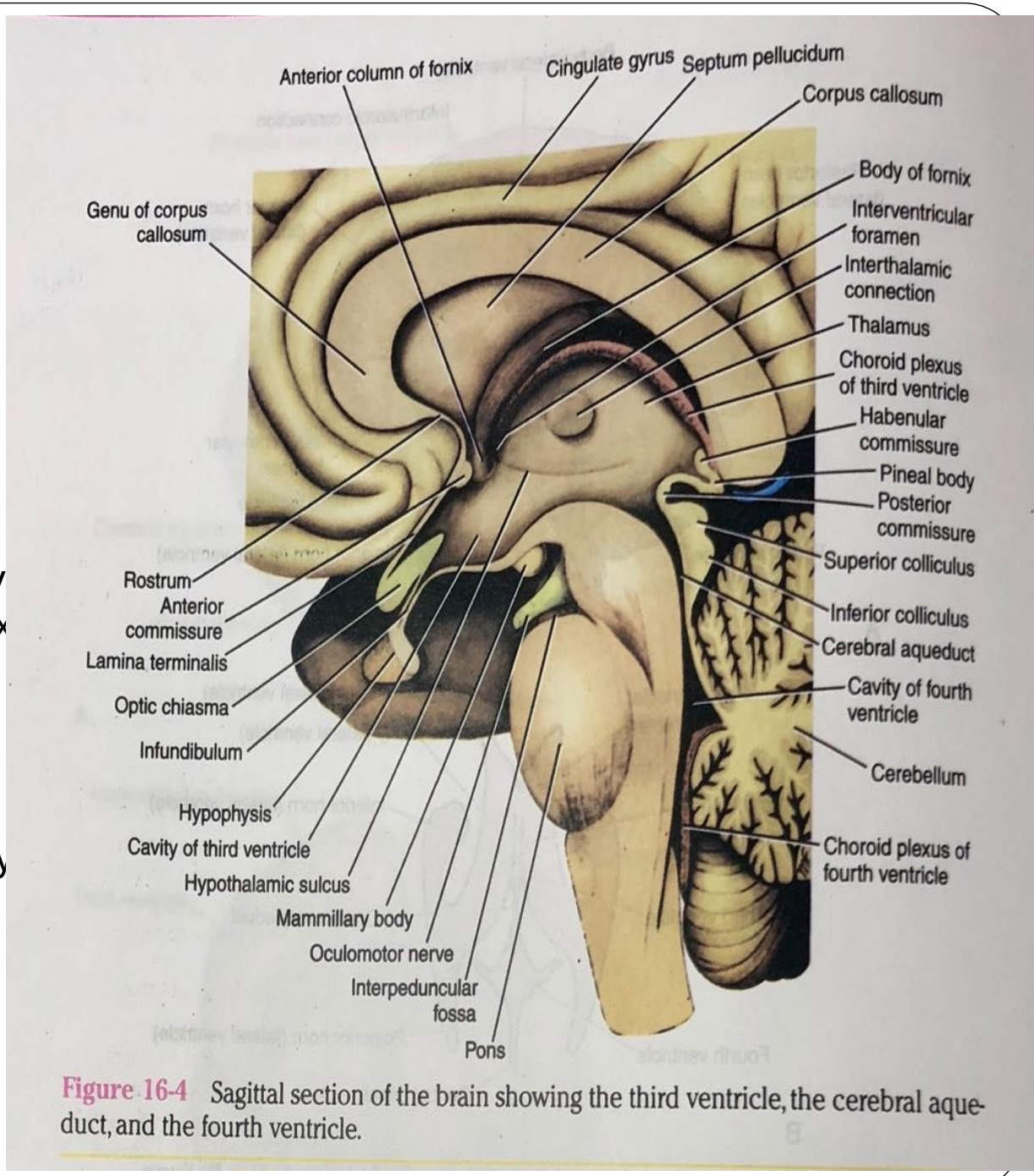


Figure 16-4 Sagittal section of the brain showing the third ventricle, the cerebral aqueduct, and the fourth ventricle.

- **Roof:**
  Formed by the ependyma that stretches across the upper limits of two thalami. (choroid plexus)

- **Floor:**
  Floor is formed by:
  - Optic chiasma
  - Infundibulum (pituitary stalk)
  - Mammillary bodies
  - Posterior perforated substance
  - Tegmentum of the mid brain

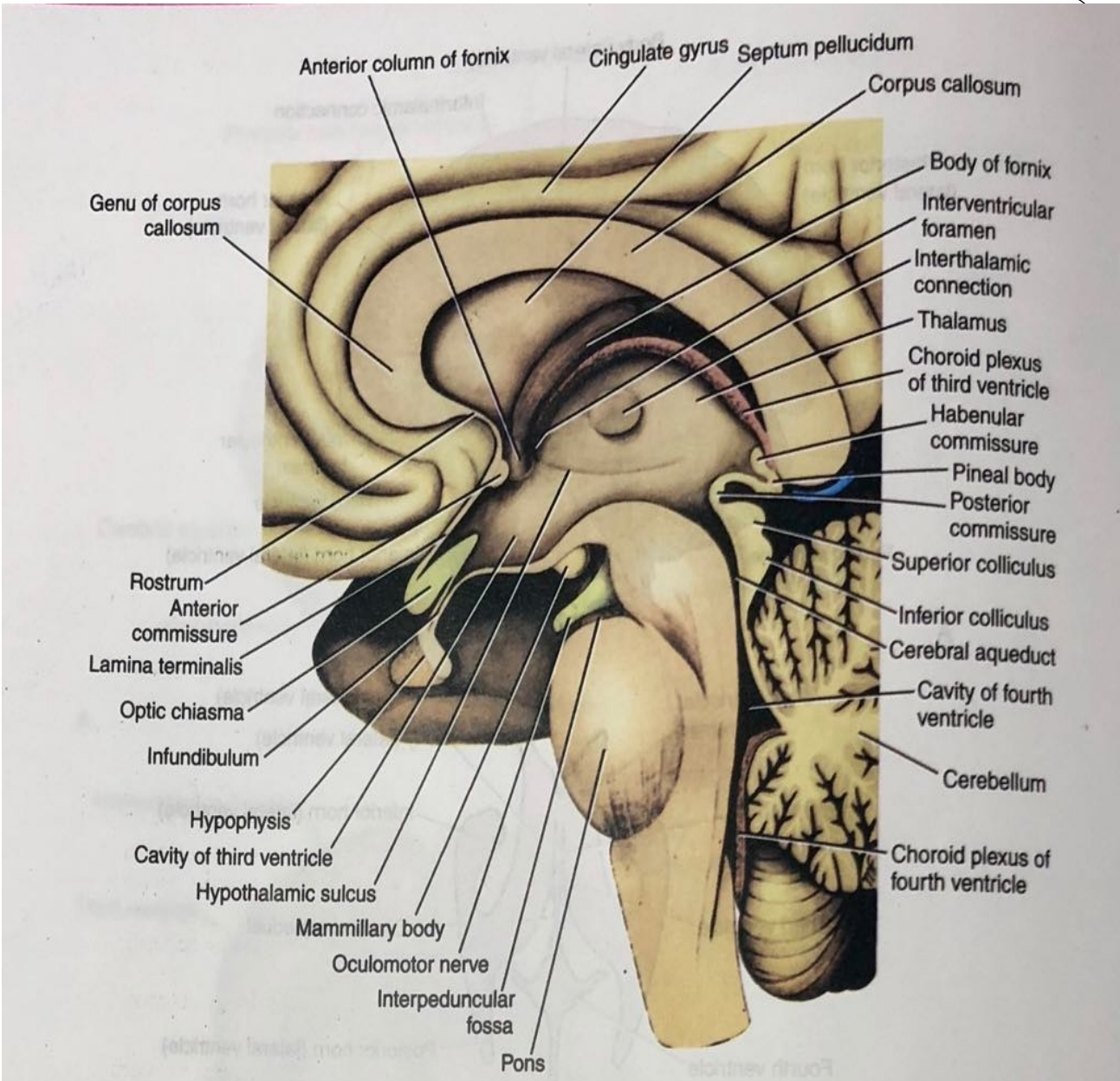


Figure 16-4 Sagittal section of the brain showing the third ventricle, the cerebral aqueduct, and the fourth ventricle.

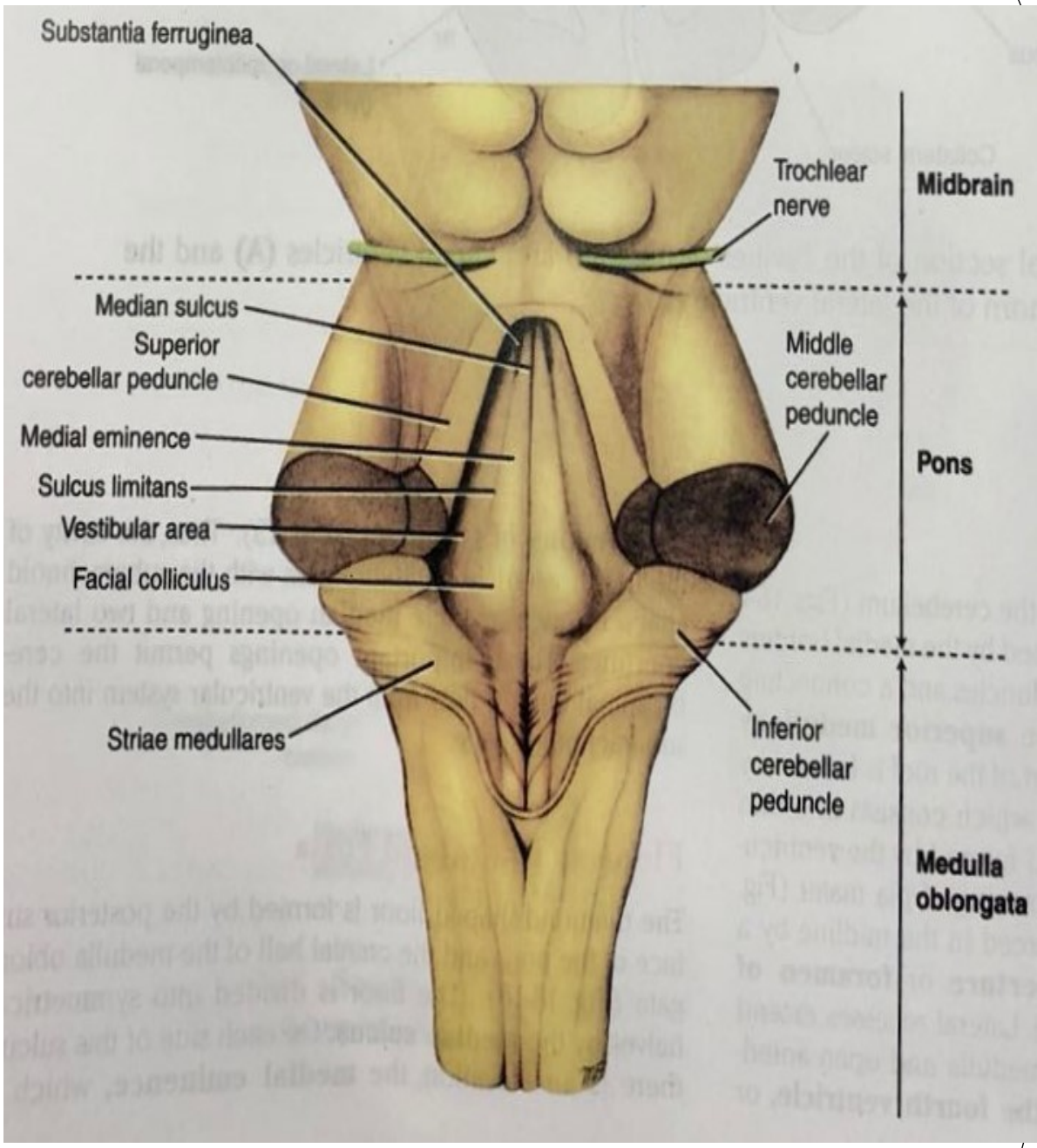
4th ventricle

- A tent-shaped cavity filled with cerebrospinal fluid.
- Situated anterior to cerebellum and posterior to pons and medulla oblongata(superior half)

BOUNDARIES:

Lateral walls:

- cranial part: by superior cerebellar peduncle
- Caudal part: by inferior cerebellar peduncle



Roof or Posterior wall:

The tent-shaped roof projects into the cerebellum

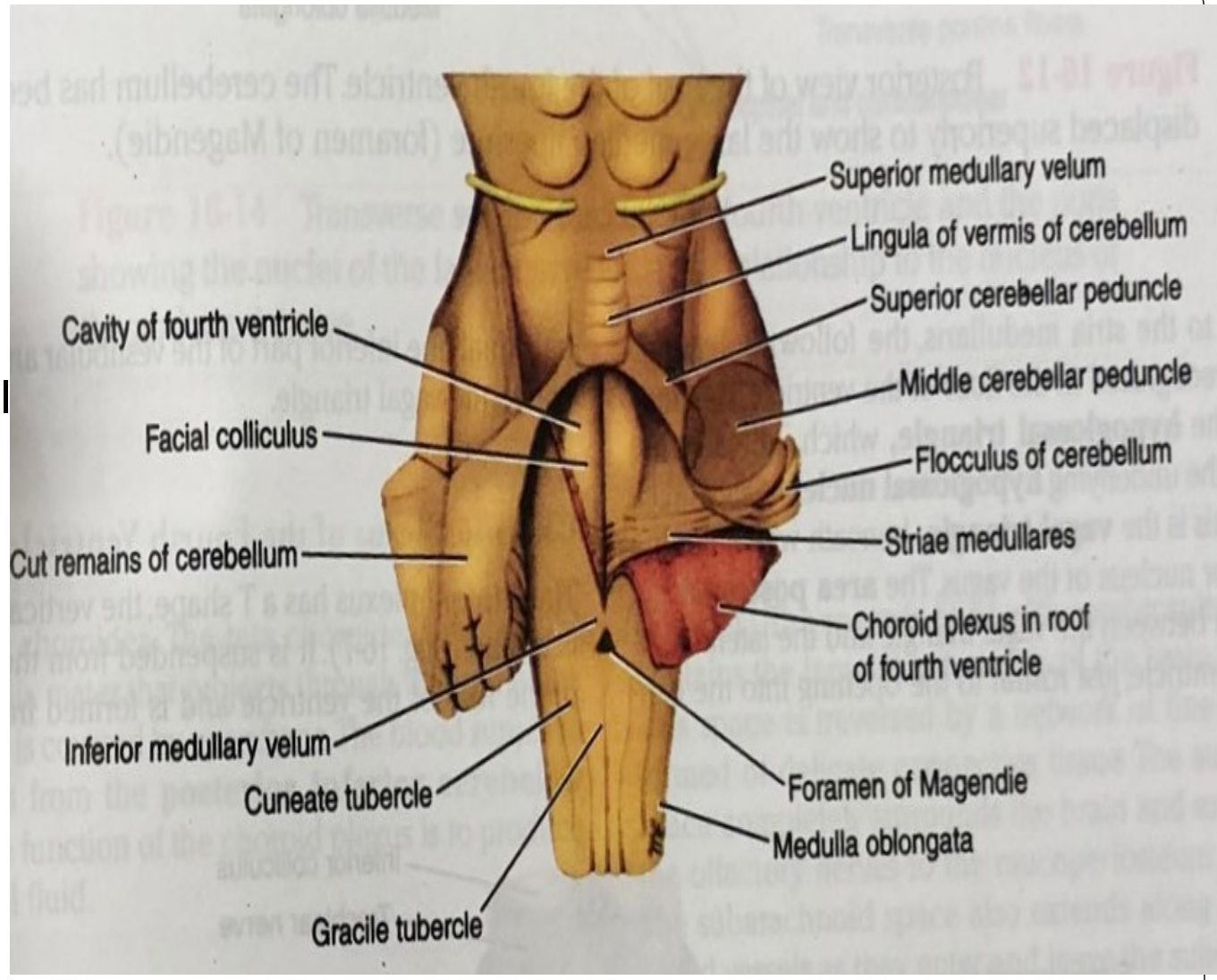
Superior part:

- 2 superior cerebellar peduncles (medial border)
- Superior medullary velum

Inferior part:

- inferior medullary velum

- **Openings in the roof:**
  - Foramen of magendie (midline)
  - 2 lateral foramina: foramina of lushka

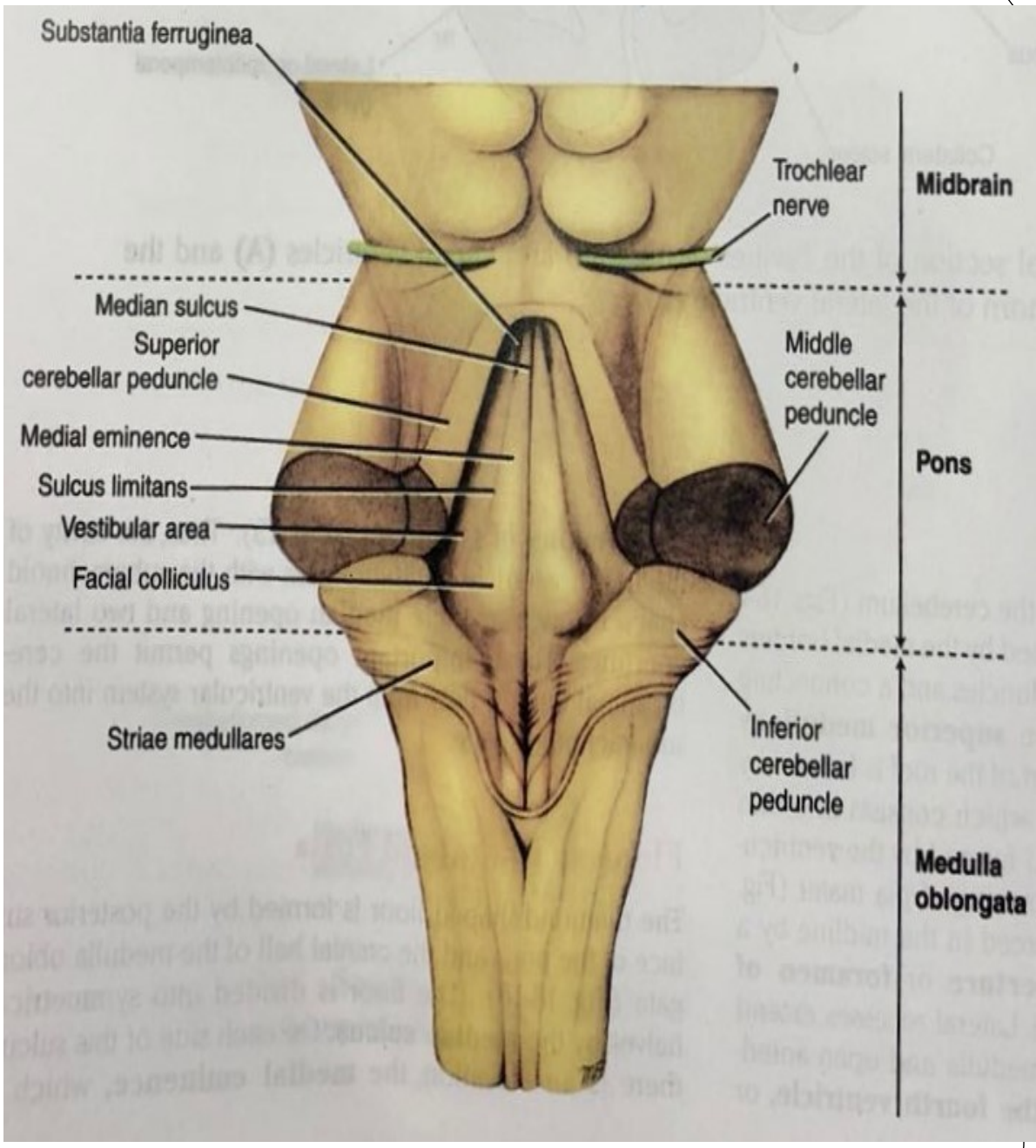


Floor or Rhomboid fossa:

The diamond-shaped floor is formed by:

- Posterior surface of the pons
- Superior half of the medulla oblongata

- The floor is divided into two halves by median sulcus
- On each side of sulcus is an elevation called median eminence
- Lateral to median eminence is sulcus limitans
- Lateral to sulcus limitans is vestibular area (vestibular nuclei is beneath it)



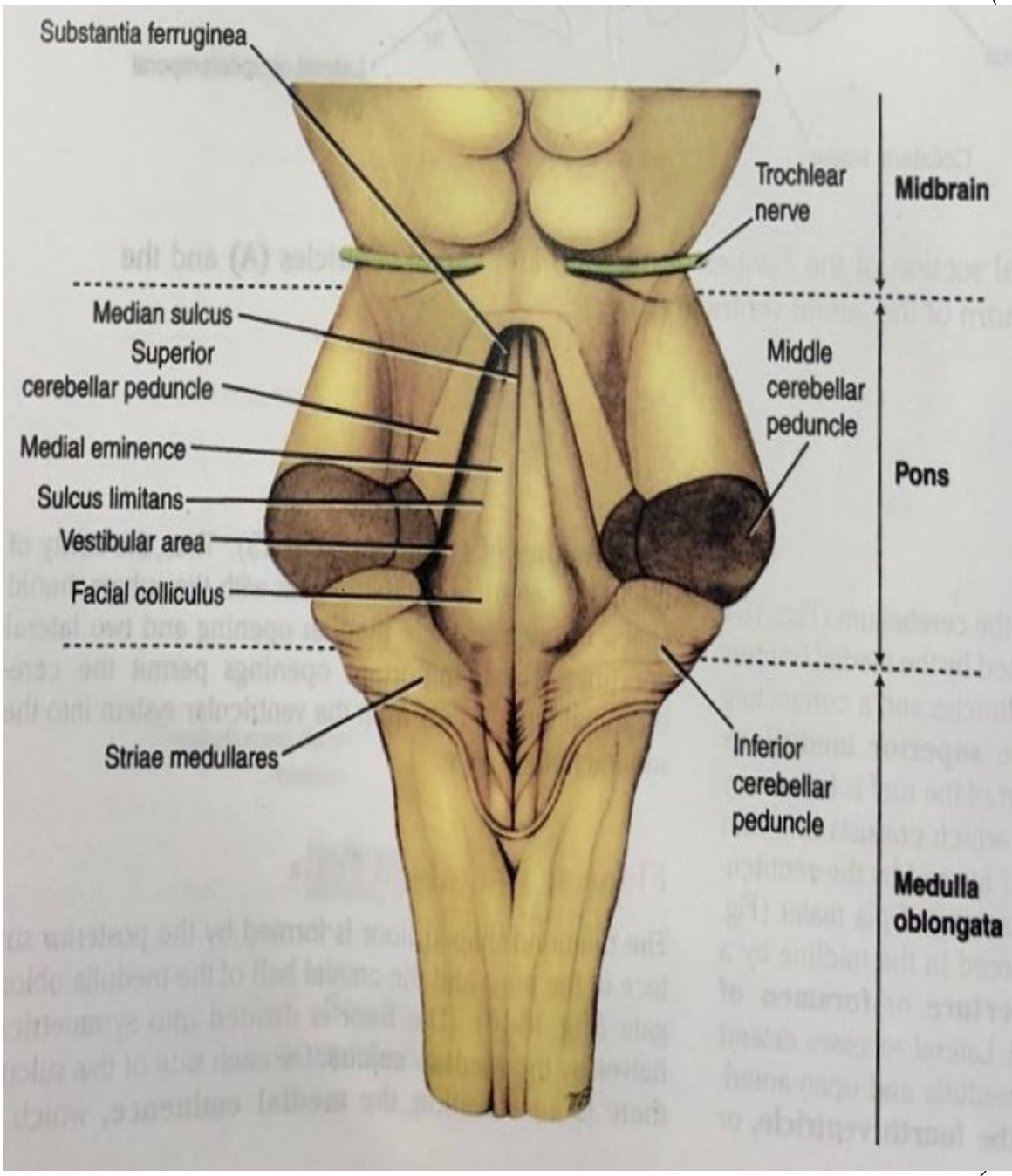
- **Nerve nuclei in the floor of 4th ventricle:**

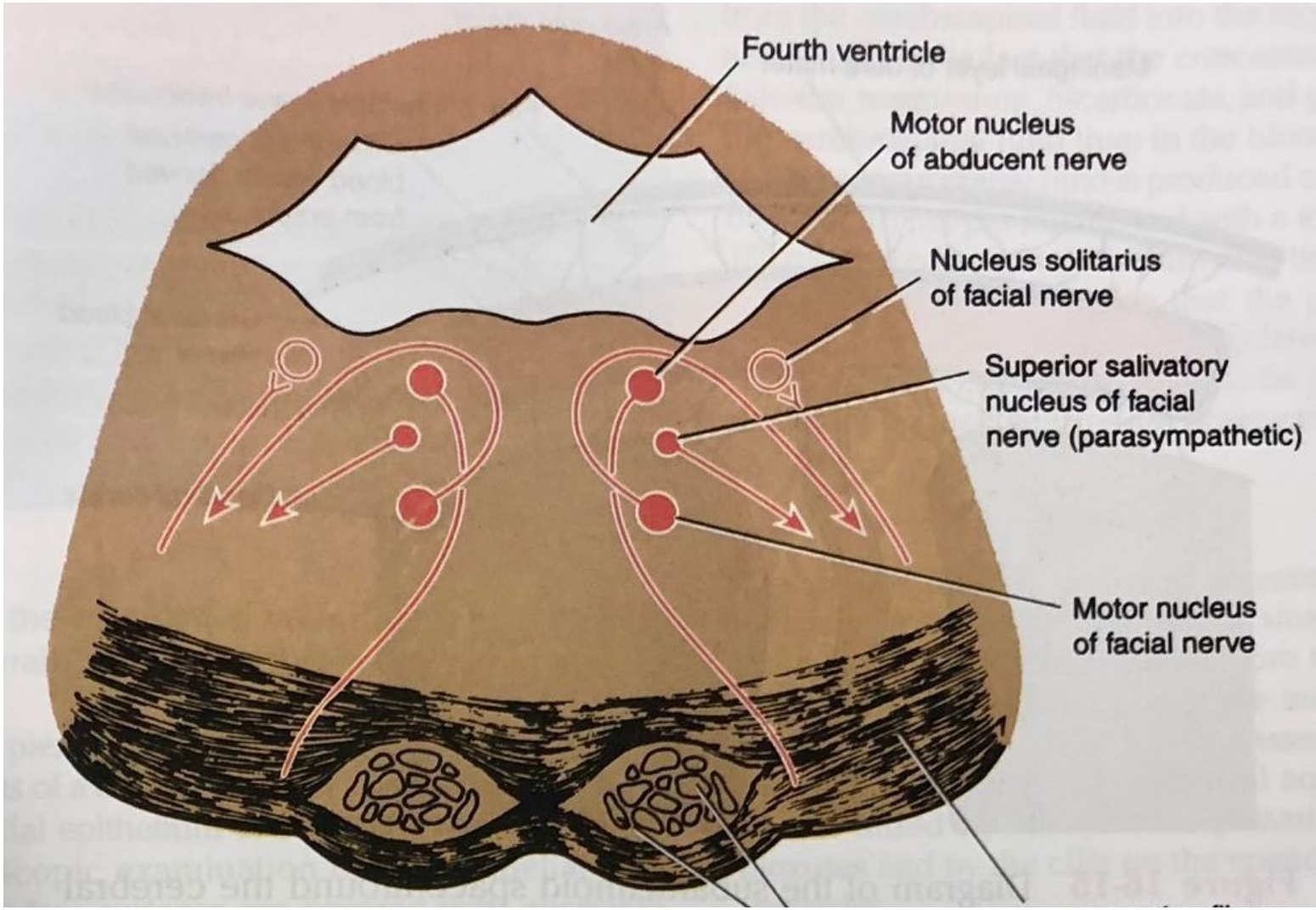
- The facial colliculus:
  - Swelling at the inferior end of the medial eminence
  - Produced by the fibres from the motor nucleus of the facial nerve looping over the Abducens nucleus.

- Substantia ferruginea:
  - Bluish area at superior end of sulcus limitans
  - Produced by cluster of nerve cells containing melanin pigment

- Stria medullaris:

  Strands of nerve fibres derived from the arcuate nuclei, emerge from the median sulcus and pass laterally over the medial eminence, and the vestibular area and enter the inferior cerebellar peduncle to reach the cerebellum.




Fourth ventricle
Motor nucleus of abducent nerve
Nucleus solitarius of facial nerve
Superior salivatory nucleus of facial nerve (parasympathetic)
Motor nucleus of facial nerve

- Inferior to stria medullaris following structures are present:
  - **Hypoglossal triangle** (having hypoglossal nucleus) –most medial
  - **Vagal triangle** (having dorsal motor nucleus of vagus) – lateral to hypoglossal triangle
  - **Area postrema** –narrow area between the vagal triangle and the lateral margin of the ventricle

Choroid plexus of ventricles

- Choroid plexus is formed by the projections of tena choroidea and is covered with the ependymal lining of the ventricular cavity.

  Tela choroidea: two-layered vascular fold of piamater situated between Fornix and Thalamus.

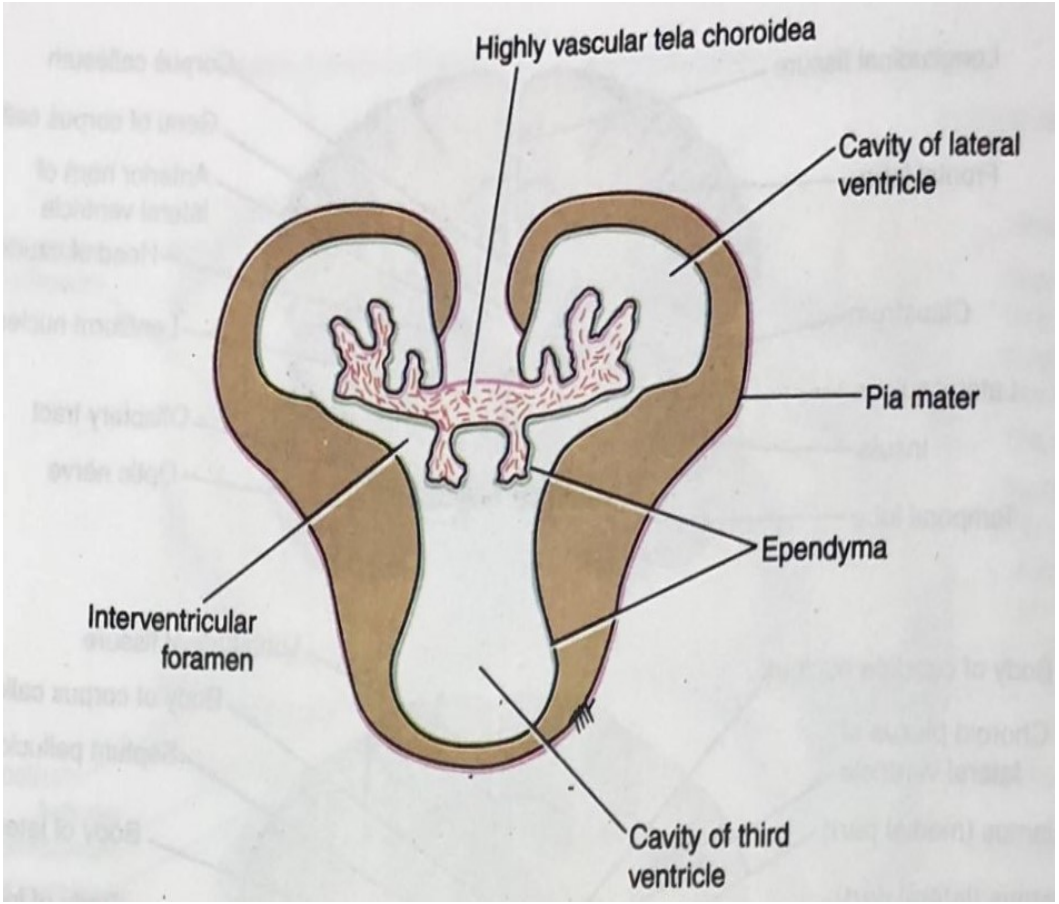
- Choroid plexus produces cerebro-spinal fluid.

- Blood supply:

  Choroidal arteries >> branches of internal carotid artery and basilar artery

- Venous drainage:

  Internal cerebral veins > great cerebral vein > sagittal sinus

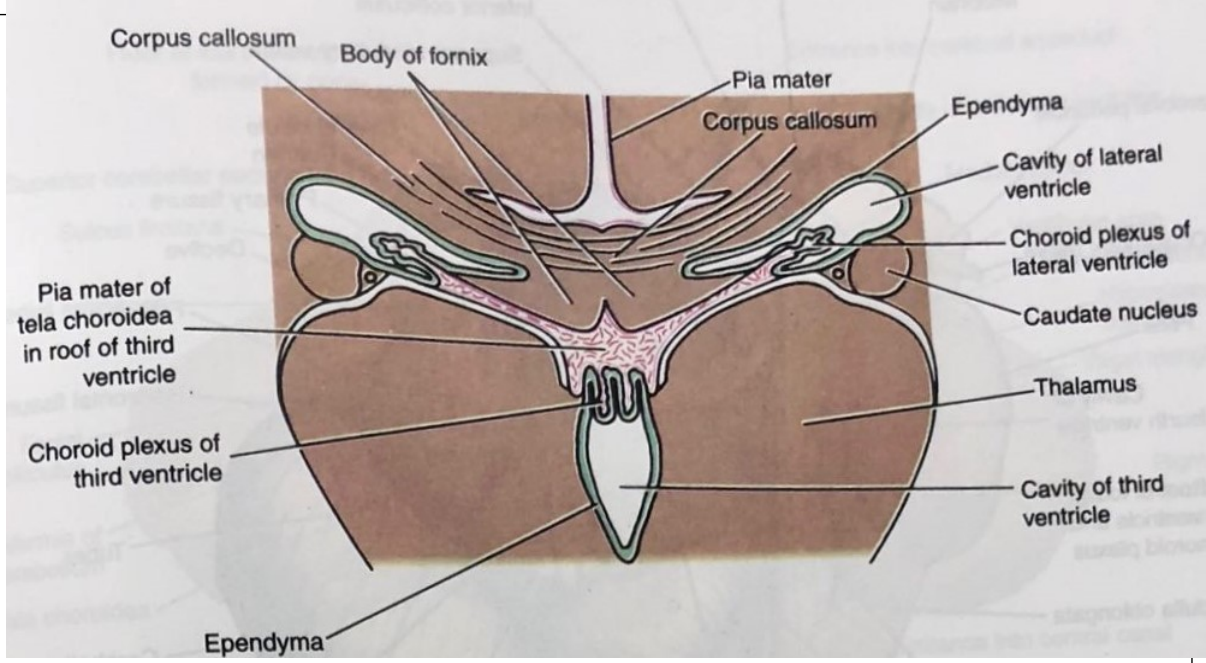


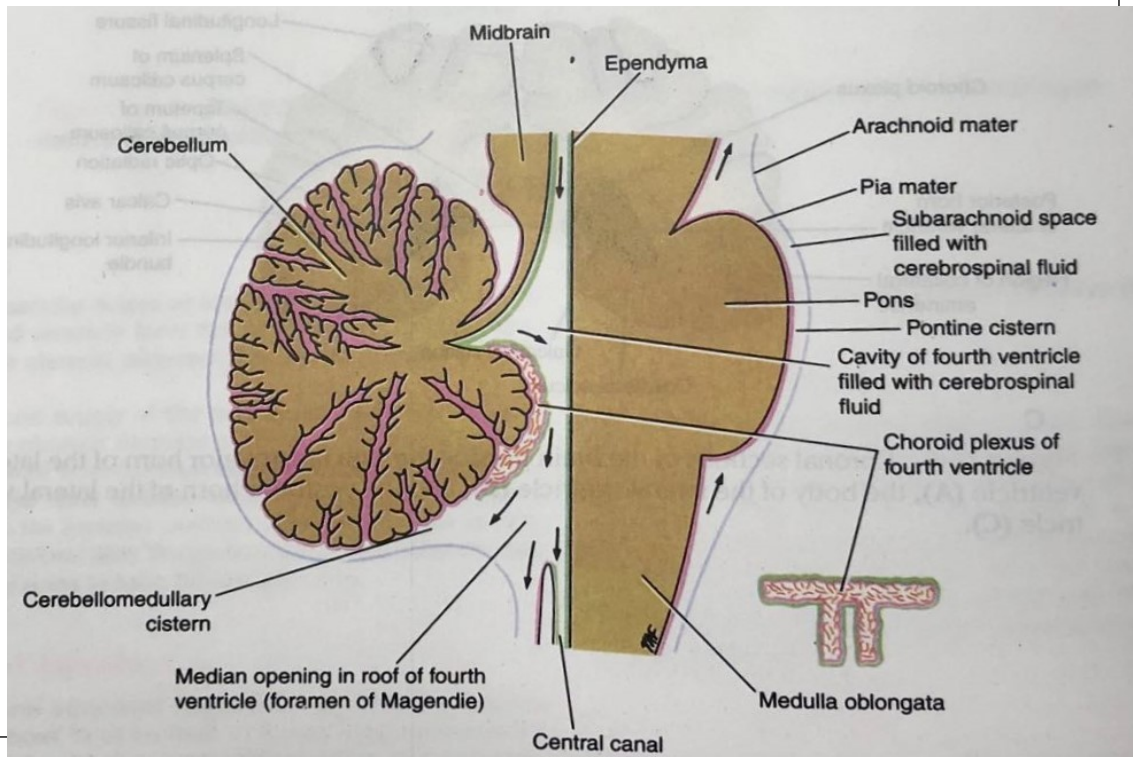
Choroid plexus

- Lateral ventricles:
  - In lateral ventricles choroid plexus projects on its medial aspect
  - At the junction of the body and inferior horn, the choroid plexus is continued into the inferior horn

- 3rd ventricle:
  - Choroid plexus passes into the interventricular foramen and is present at the top of the 3rd ventricle forming its roof.
  - Two vascular ridges hang from the floor

- 4th ventricle:
  - The choroid plexus is a T shape
  - The vertical part of the T is double
  - It is suspended from the inferior half of the roof of ventricle





Ciculation of CSF

Lateral ventricles

↓

The foramen of Monro

↓

3rd ventricle

↓

Cerebral aqueduct

↓

4th ventricle → Central canal of spinal cord

↓

Foramen of magendie and lushka

↓

Sub-arachnoid space

↓

Arachnoid granulations (groups of arachnoid villi)

↓

Superior sagittal sinus

THANK YOU